

Designing Constitutional Review Institutions by Simulation

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Abstract

Constitutional review institutions face a recurring design problem: they need enough independence to preserve rights and legal continuity, but enough accountability to avoid becoming unreviewable partisan veto points. This paper introduces a computational design-search model for comparing constitutional-review institutions under common docket and political assumptions. The model represents appointment systems, court size, tenure, removal, recusal, emergency procedure, voting thresholds, panel and en banc review, cross-checking courts, constitutional councils, legislative overrides, admissibility filters, and post-decision political response. The current campaign evaluates fourteen institutional designs across twenty ordinary and adversarial assumption cases. A filed-universe intake model routes cases through review paths and admission filters before merits review, and the reports keep direct outputs such as rights-claimant success, public confidence, lower-court compliance, and doctrinal depth separate from derived indices. The results do not identify an optimal court. They show how design bundles shift tradeoffs among legal stability, rights protection, emergency-docket abuse, partisan alignment, legitimacy, constitutional conflict, and democratic responsiveness.

Keywords: constitutional review, judicial politics, institutional design, simulation, courts

1 Design Question

The central question is not whether one court design is ideal in isolation. It is how bundles of constitutional-review rules alter institutional tradeoffs when they face the same filed universe, the same political environment, and the same strategic pressures. A court can be independent while losing democratic legitimacy, responsive while failing rights protection, stable while entrenching poor doctrine, or active while creating constitutional conflict. The simulator is built to expose those tensions rather than collapse them into a single institutional label.

This makes the paper a computational institutional-design exercise for judicial politics and comparative public law. The simulator is not intended to forecast any real supreme court. It is a design-search scaffold for asking how appointment regularity, emergency procedure, admission filtering, voting thresholds, override rules, council warnings, cross-court checks, and post-decision compliance strategies change the distribution of outcomes. That framing is especially important because many proposed court reforms are defended one rule at a time, even though their effects depend on the surrounding institutional bundle.

2 Model

Each generated world contains a pool of potential justices, a filed universe of constitutional disputes, and a legislative-output profile. Filed cases vary by legal ambiguity, rights burden, democratic mandate, partisan salience, emergency pressure, requested emergency relief, legislative quality, constitutional-conflict potential, public attention, override pressure, lower-court conflict,

lower-court error risk, and certiorari pressure. The docket separates facial challenges, as-applied challenges, election disputes, emergency stay applications, executive-power disputes, administrative-law challenges, structural cases, economic-regulation cases, and rights claims. Legal-domain profiles shift baseline case attributes so rights, election, executive-power, administrative, structural, and economic-regulation cases do not behave as interchangeable labels.

A review-path layer distinguishes paid certiorari, IFP certiorari, discretionary certiorari, direct constitutional complaint, amparo, filtered QPC-style referral, abstract ex ante and ex post review, concrete referral, emergency application, interbranch dispute, and electoral review. Admission filtering turns filed disputes into screened-out matters or merits transfers using SG/CVSG-style signal, amicus intensity, split maturity, relists, specialist counsel, vehicle-quality defects, and conditional reversal probability. There is no preliminary fixed-size top-case selector before admission; the modeled merits docket is an output of the intake process.

Each scenario supplies a court design. The design selects reviewers, applies recusal rules, routes cases to full-court or panel review, determines emergency relief procedure, applies voting thresholds, optionally invokes cross-checking or council review, and optionally allows legislative or popular override. Emergency applications carry applicant type, application class, requested relief, response request, full-court referral, status-quo effect, public disagreement risk, and merits follow-through. Strategic actors choose whether to comply, evade, reenact, flood emergency applications, organize override campaigns, or pressure appointments. Their response is split into formal legal response and practical implementation response, including repeal, replacement, narrowed reenactment, weak-form override, amendment, court-curbing, delay, administrative substitution, symbolic compliance, bureaucratic resistance, and open noncompliance.

3 Calibration and Validation

The calibration layer is deliberately framed as a plausibility guardrail rather than as a claim of full validation. Normalized source observations are drawn from the Supreme Court Database, Harvard Law Review Supreme Court statistics, the Supreme Court Shadow Docket Database, Black and Epstein’s recusal data, and Deep Research synthesis tables (Supreme Court Database, 2025; Harvard Law Review, 2025; Kastellec and Taboni, 2025; Black and Epstein, 2005; Supreme Court Simulator Deep Research Synthesis, 2026). The current calibration run passes all seventeen guardrails for docket mix, invalidation, recusal frequency, emergency-docket abuse, split legal stability, partisan alignment, override attempts, and admissibility-aware merits transfer.

The model distinguishes three categories that should not be blurred. Observed inputs anchor source ranges or qualitative rules. Coding decisions translate legal institutions into parameterized mechanisms. Simulated outputs are exploratory comparisons under those mechanisms. Several quantities remain intentionally loose: SCOTUSblog significant-application counts are not full emergency denominators; raw cross-national invalidation rates are not directly comparable without institutional context; and source tables describing appointment actors or court structure are not treated as direct measures of independence. The generated source-range appendix reports raw source minima, percentiles, medians, maxima, terms, and source keys, while the campaign reports preserve deterministic seeds and scenario definitions for reproduction.

4 Metrics

The headline metrics are legal stability, rights protection, partisan alignment, shadow-docket abuse, legitimacy, reversal rate, constitutional conflict, and democratic responsiveness. Legal stability is

decomposed into precedent stability, statutory stability, and interbranch compliance. Emergency legitimacy risk is reported separately from shadow-docket abuse, so opacity and emergency procedure legitimacy are not collapsed into the same quantity as abusive merits-like relief. Rights-claimant success is reported both as an aggregate direct output and by legal domain for rights, structural, election, executive-power, administrative-law, and economic-regulation cases.

The reported directional score is only a reading aid. It averages a stability/rights score, a legitimacy/control score, rights-claimant success, elite acceptance, and administrative feasibility. Direct outputs such as rights-claimant success, doctrinal depth, remedial breadth, lower-court compliance, elite acceptance, public confidence, and coalition fragmentation are reported separately and should not be treated as interchangeable components of one welfare function.

	Rights	Struct.	Elect.	Exec.	Admin.	Econ.
CUR	0.27	0.17	0.53	0.13	0.11	0.08
TERM	0.26	0.17	0.43	0.10	0.10	0.09
EMERG	0.26	0.15	0.46	0.10	0.10	0.07
COMM	0.27	0.17	0.43	0.09	0.10	0.08
S60	0.20	0.14	0.32	0.06	0.08	0.06
PANEL	0.26	0.17	0.42	0.10	0.10	0.09
SENATE	0.17	0.10	0.27	0.04	0.06	0.04
CHECK	0.16	0.11	0.24	0.06	0.05	0.05

Domain-specific claimant-success rates; darker cells are higher.

Figure 1: Domain-specific rights-claimant success for selected institutional designs. The aggregate rights-protection score can hide domain shifts, especially for election, structural, executive-power, administrative-law, and economic-regulation cases.

5 Campaign Results

The current campaign compares fourteen institutional scenarios across twenty assumption cases: the original broad cases, sensitivity sweeps for appointment capture, emergency pressure, rights risk, and democratic mandate, plus adversarial stress cases for appointment timing, emergency-application flooding, override evasion, recusal pressure, and court-expansion retaliation. The reported run used 80 runs per case, 64 filed matters per run, seed 20260501, and an imported legislative-output profile from a separate legislative simulator report contract.

Table 1 shows that term regularization leads the current directional score by combining low shadow-docket abuse, moderate emergency legitimacy risk, and relatively low conflict. The no-emergency-relief-without-merits design remains the strongest emergency-docket reform on shadow abuse, emergency legitimacy risk, and public confidence, though it pays administrative cost through merits acceleration. The stylized current U.S.-like court has high rights protection in this run, but much higher shadow-docket abuse and emergency legitimacy risk. Cross-checking and dual-court designs reduce partisan alignment, but the current disagreement-resolution model increases constitutional conflict and administrative cost. The comparative 16-seat constitutional-senate

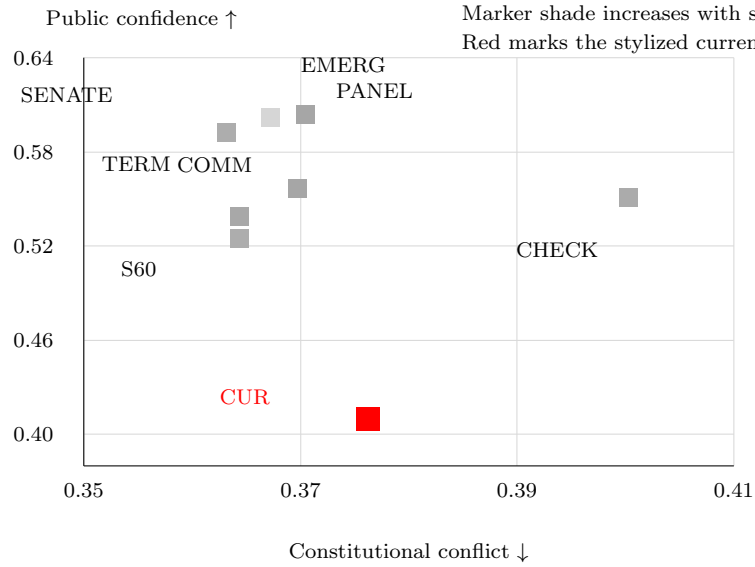


Figure 2: Public-confidence and constitutional-conflict tradeoff for selected institutional designs. Marker shade increases with shadow-docket abuse, so the figure separates legitimacy and conflict from emergency-docket opacity instead of folding them into one score.

Table 1: Selected v2 campaign averages

Scenario	Score	Admit	Rights	Shadow	Emerg. risk	Public	Conflict
18-year staggered terms	0.588	0.560	0.613	0.068	0.190	0.539	0.364
No emergency relief without merits review	0.585	0.582	0.617	0.004	0.187	0.602	0.367
Stylized current U.S.-like court	0.584	0.560	0.618	0.235	0.255	0.410	0.376
Nonpartisan commission	0.584	0.578	0.616	0.070	0.197	0.557	0.370
60 percent invalidation threshold	0.583	0.561	0.602	0.061	0.200	0.525	0.364
Panel with en banc correction	0.581	0.582	0.614	0.071	0.196	0.604	0.371
Comparative 16-seat constitutional senates	0.570	0.562	0.597	0.062	0.200	0.593	0.363
Cross-checking constitutional court	0.551	0.577	0.595	0.063	0.205	0.551	0.400

scenario confirms that even court sizes and chamber-like structures can be modeled, but it does not automatically dominate simpler reforms.

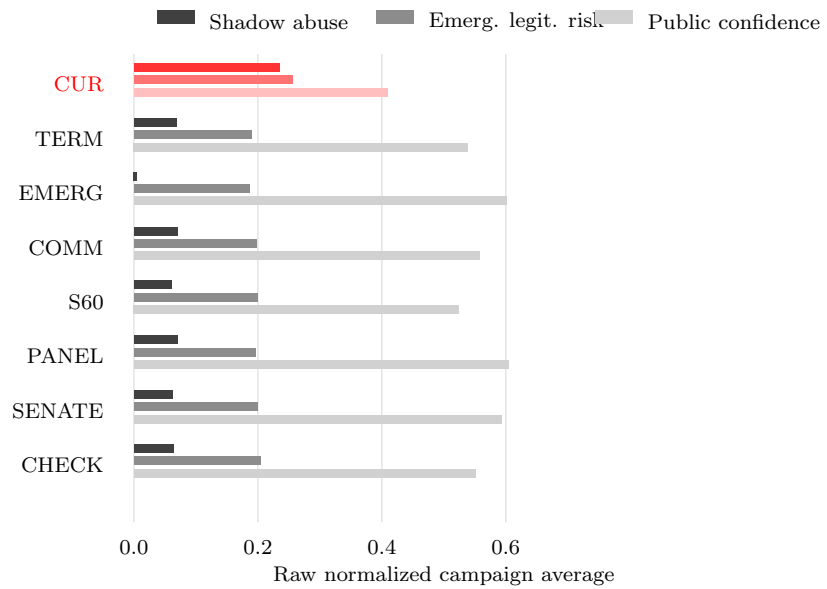


Figure 3: Emergency-docket and public-confidence profile for selected institutional designs. The comparison highlights why emergency procedure is modeled directly: designs that are similar on rights protection and legal stability can differ sharply on shadow-docket abuse, emergency legitimacy risk, and public confidence.

6 Limitations

The model is strongest as a disciplined comparison of institutional mechanisms, not as a predictive empirical model. It does not estimate causal effects from observed constitutional systems. It uses source-backed ranges, comparative institutional coding, and deterministic scenario campaigns to identify where rule bundles create tradeoffs that would otherwise be difficult to see. The strongest next validation steps are to automate raw-data refresh for SCDB, Harvard Law Review statistics, shadow-docket data, and recusal data; convert named priors into sampled prior distributions; add actor learning across repeated outcomes; calibrate review-path-specific admission denominators against raw certiorari, amparo, constitutional complaint, QPC, abstract-review, and concrete-referral data; and benchmark the legislative import contract against raw legislative simulator runs rather than only aggregate campaign CSVs.

7 Data Availability Statement

The simulator, normalized calibration tables, deterministic seeds, scenario definitions, and generated reports are maintained as a reproducible project. Public repository identifiers are withheld in this anonymous review manuscript. The intended replication package includes source code, build instructions, normalized input data, generated campaign artifacts, and documentation distinguishing observed inputs, coding choices, model assumptions, and simulated outputs.

8 Funding Statement

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9 Competing Interests

The author(s) declare none.

A Institutional Parameter Provenance

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Table 2: Illustrative provenance map for modeled institutional features

Feature family	Model representation	Provenance status
Appointment and tenure	Appointment method, confirmation threshold, vacancy deadlock risk, term renewability, retirement age, replacement timing	Comparative institutional coding and scenario assumptions
Review access	Access path, petition type, admission score, merits transfer, screen-out	Deep Research synthesis tables and modeled admission rules
Emergency procedure	Application class, relief request, response request, full-court referral, reason giving, status-quo effect, merits follow-through	Shadow-docket source tables plus scenario rules
Decision rule	Voting threshold, panel/en banc routing, quorum failure, recusal consequence, per-remedy thresholds	Court-design coding and institutional assumptions
Post-decision politics	Override attempt, rights carveout, formal response, practical implementation, open defiance	Comparative public-law coding and strategic actor assumptions
Outputs	Stability, rights protection, claimant success, legitimacy, conflict, responsiveness	Simulated outcomes with calibration guardrails

Supreme Court Simulator Deep Research Synthesis (2026). Comparative constitutional review calibration and implementation notes. Local Deep Research report imported as normalized synthesis CSV tables under data/calibration/supreme-court-synthesis.